UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Shortest Path Problem

Shortest Path

- Let ***G*** be a weighted graph.
- Let ***u*** and ***v*** be two vertices in ***G***, and let ***P*** be a path in ***G*** from ***u*** to ***v***.
- The length of path ***P***, written ***L(P)***, is the sum of the weights of all the edges on path ***P***.
- A **shortest path** from a vertex to another vertex is a path with the shortest length between the vertices.

Dijkstra's Shortest Path Algorithm

- 1. $S := \emptyset$
- 2. $N := V$
- 3. For all vertices, $u \in V$, $u \neq a$, $L(u) := \infty$
- 4. $L(a) := 0$

http://en.wikipedia.org/wiki/Dijkstra%27s_algorithm

- 5. While $z \notin S$ do,

  5.a :Let $v \in N$ be such that

  $$L(v)=\min\{L(u) \mid u \in N\}$$

  5.b : $S := S \in \{v\}$

  5.c : $N := N - \{v\}$

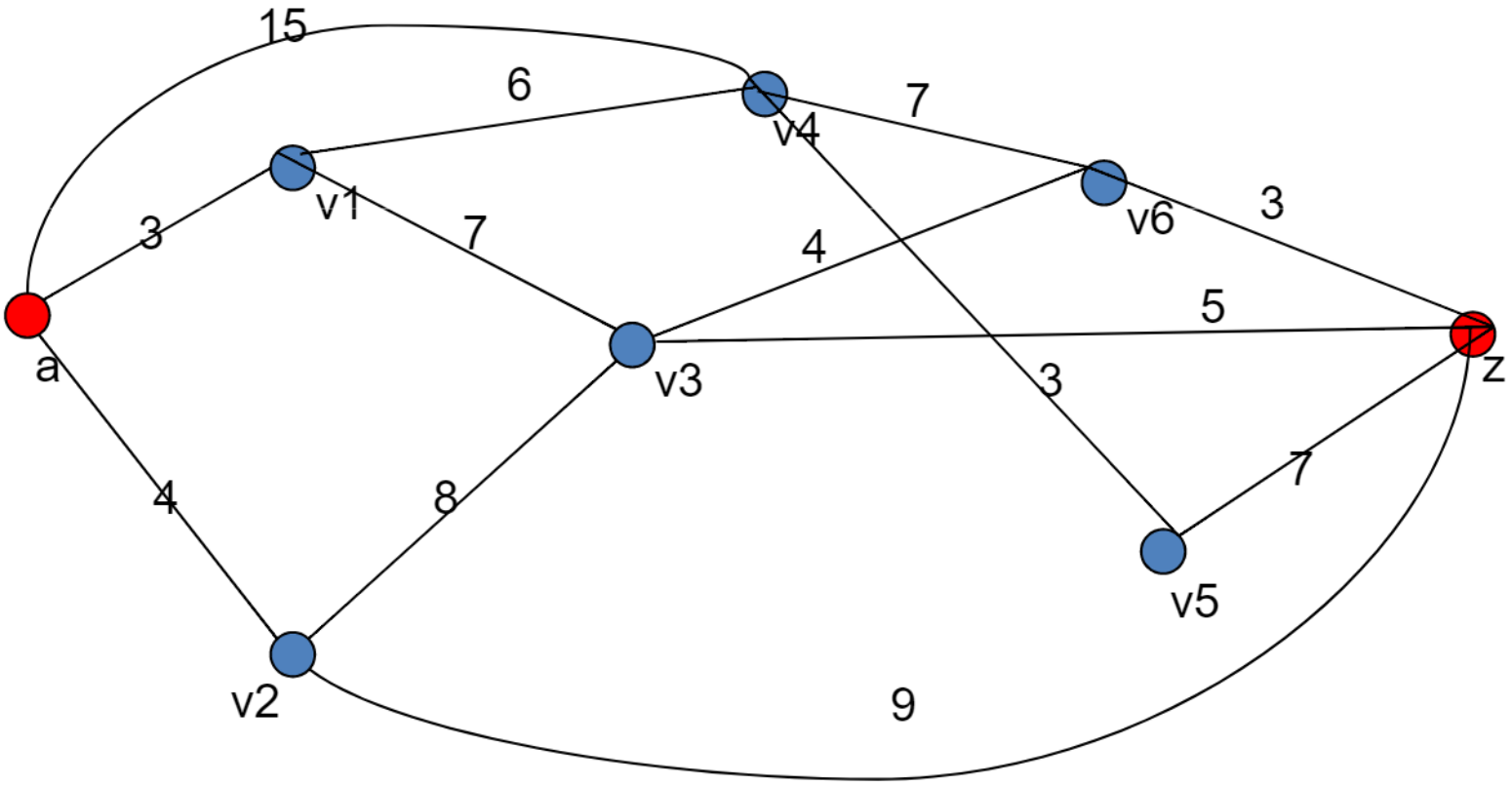
  5.d : For all $w \in N$ such that there is an edge from v to w

  5.d.1: If $L(v)+W[v,w] < L(w)$ then

  $$L(w)=L(v)+W[v,w]$$

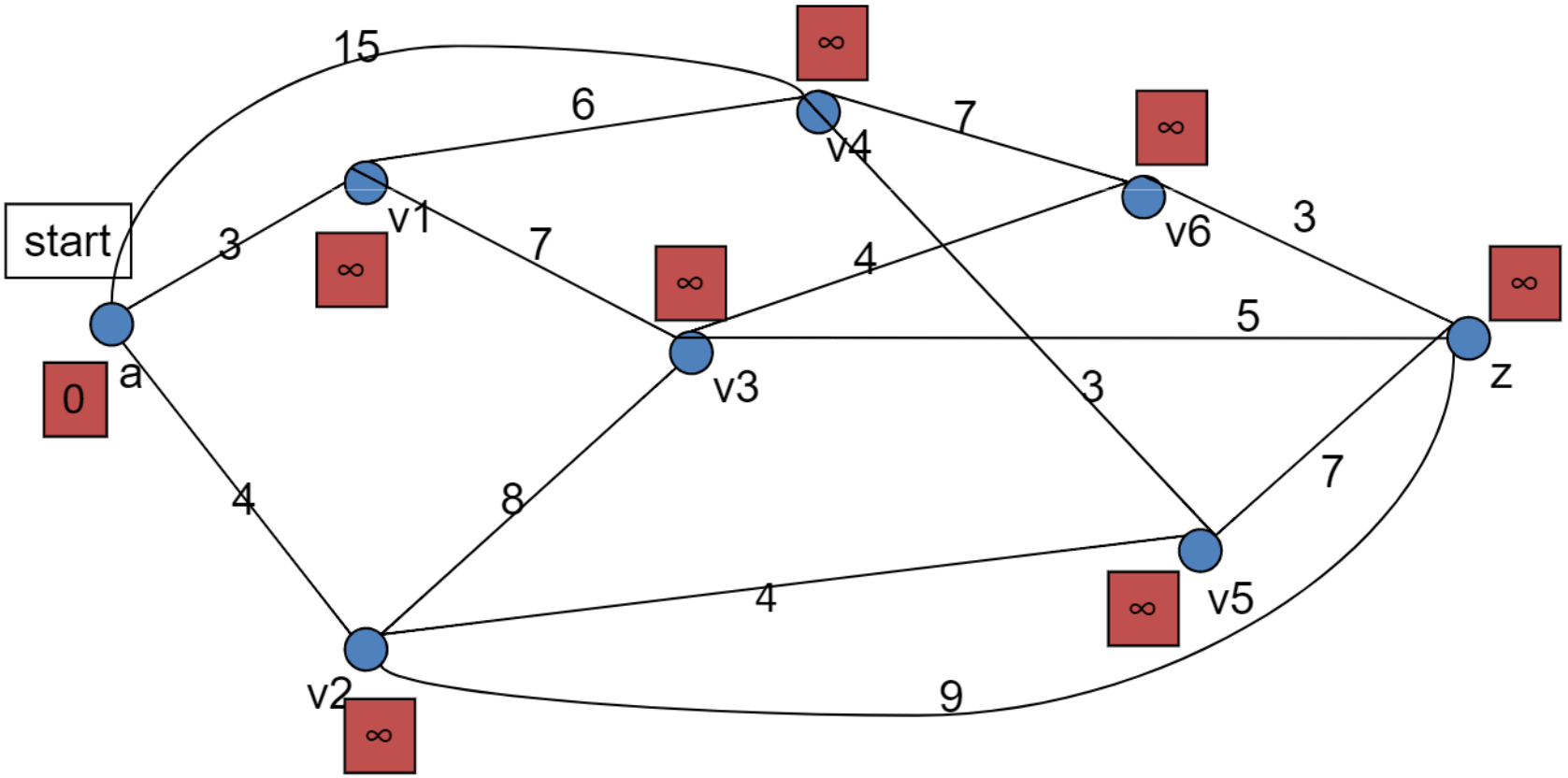
Example

What is the shortest path from **a** to **z**?



$S=\emptyset$

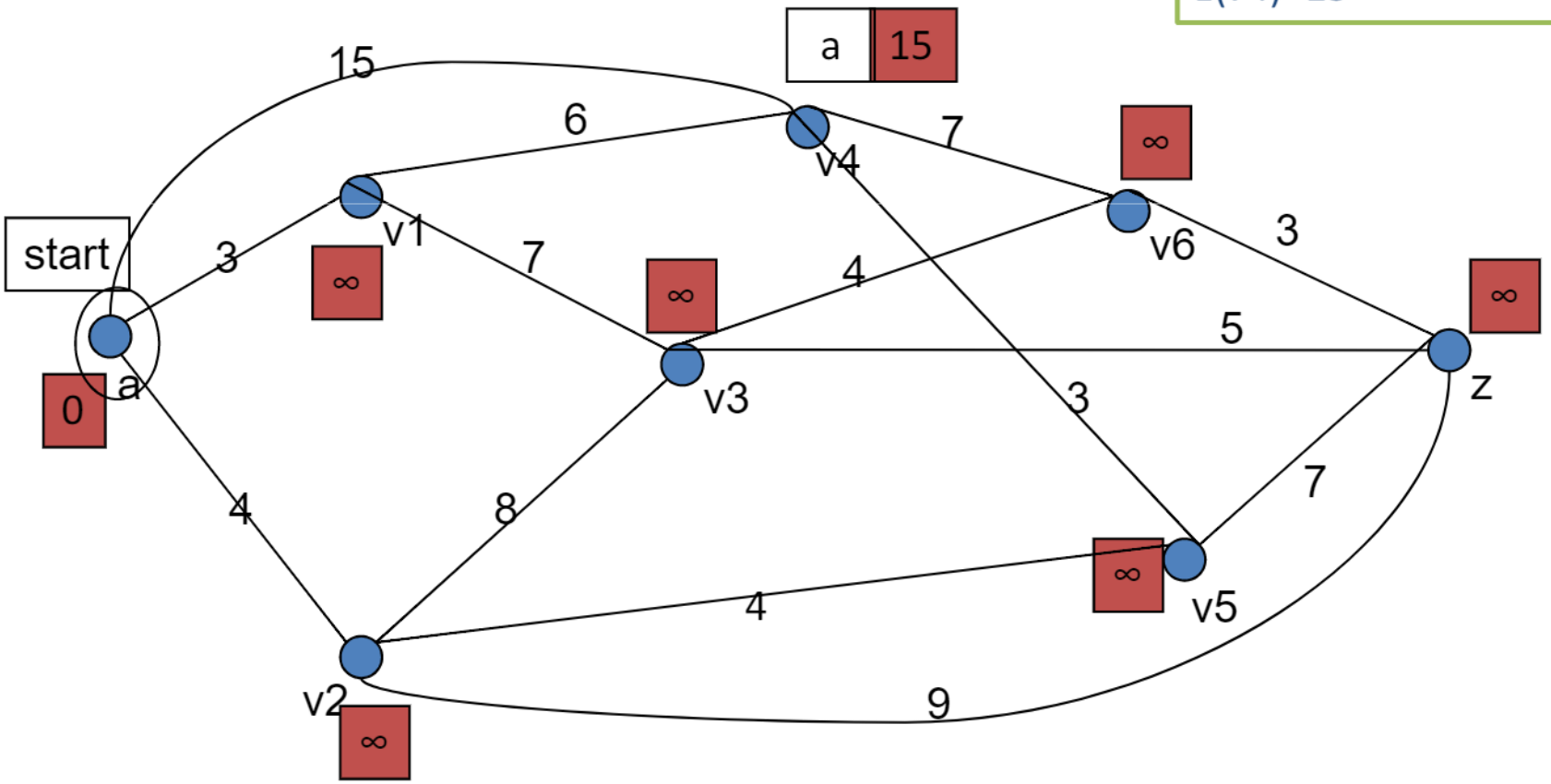
$N=\{a,v1,v2,v3,v4,v5,v6,z\}$



$S=\{a\}$

$N=\{v1,v2,v3,v4,v5,v6,z\}$

$L(a)+W[a,v4] < L(v4)$
$0+15=15 < \infty$
$L(v4)=15$



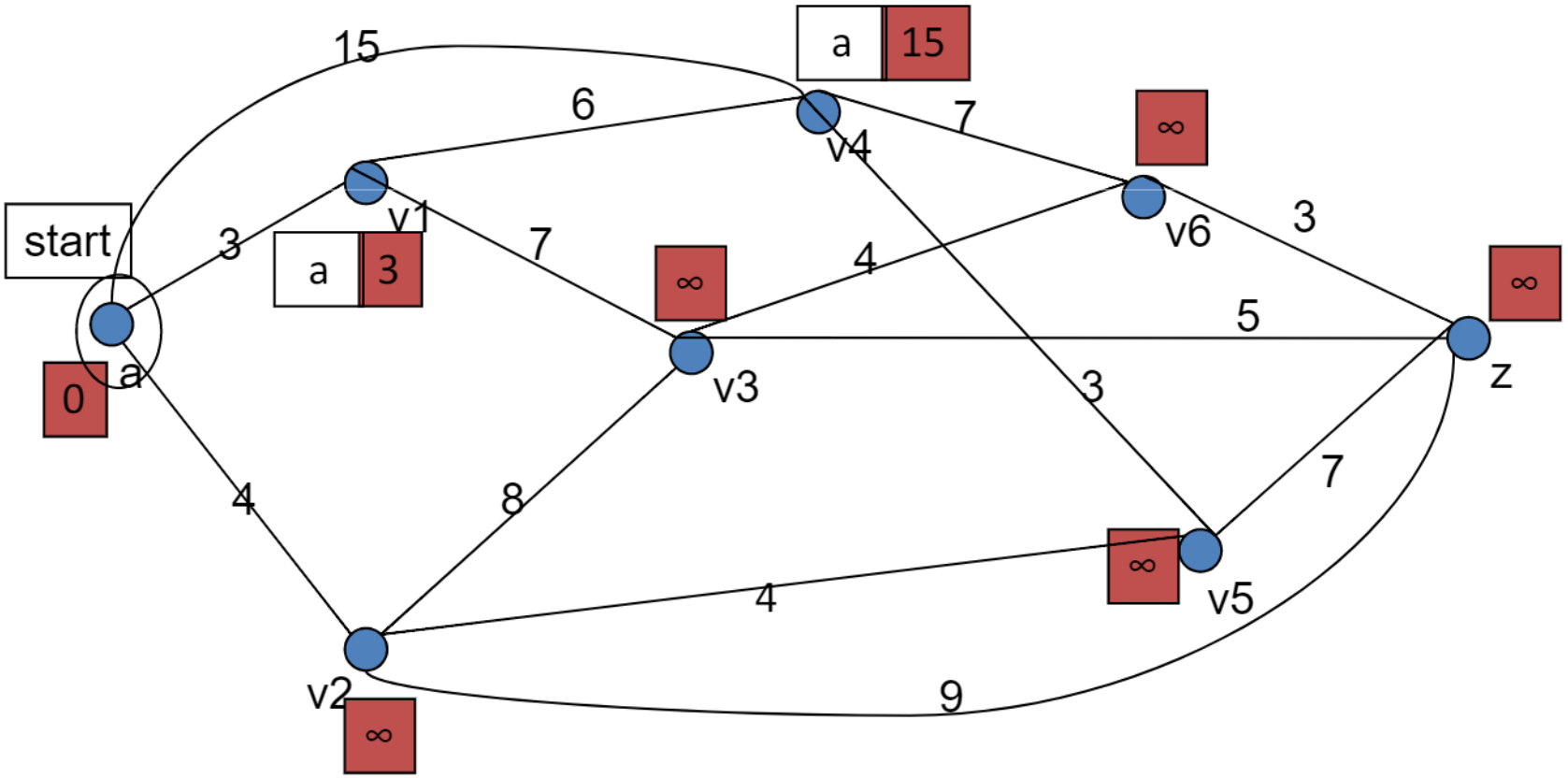
$S=\{a\}$

$N=\{v1,v2,v3,v4,v5,v6,z\}$

$L(a)+W[a,v1] < L(v1)$

$0+3=3 < \infty$

$L(v1)=3$



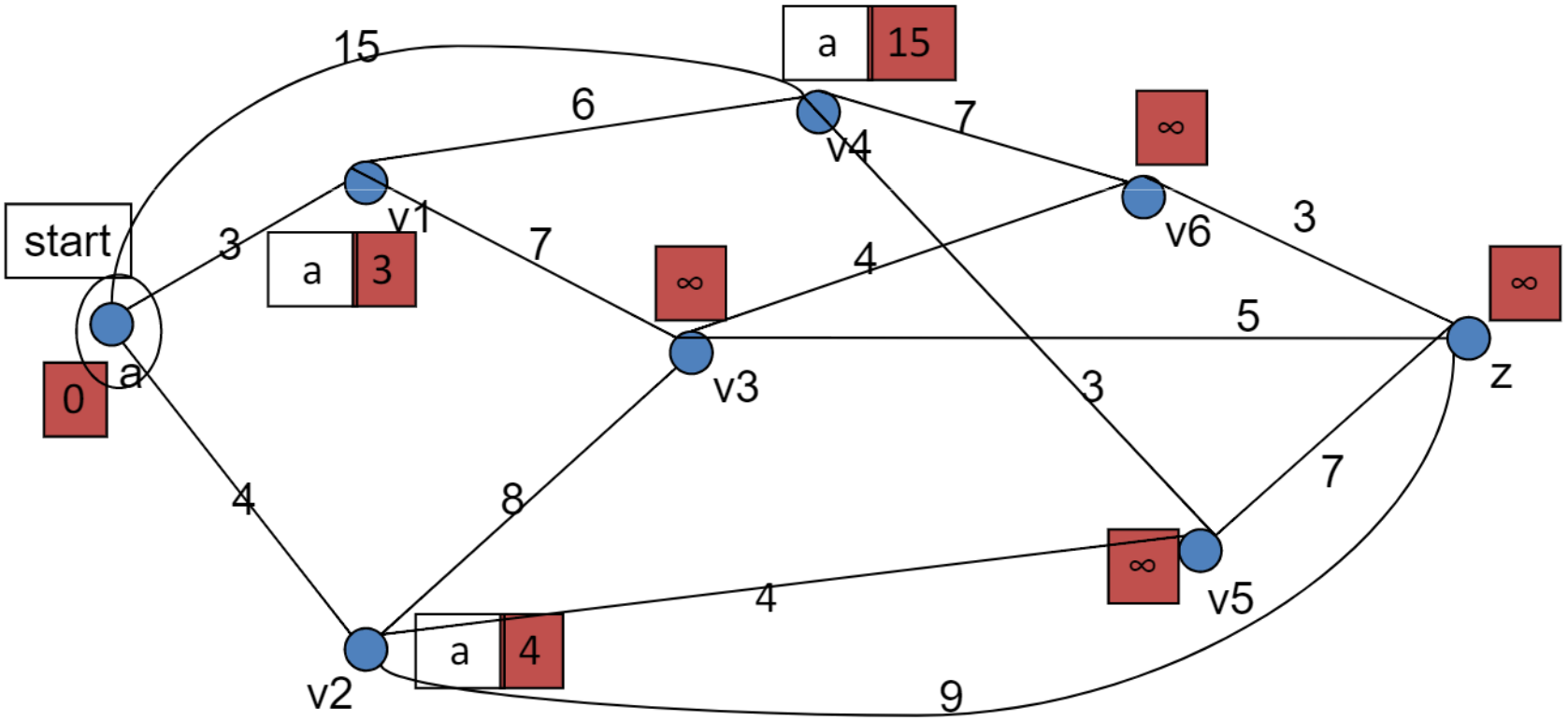
$S=\{a\}$

$N=\{v1,v2,v3,v4,v5,v6,z\}$

$L(a)+W[a,v2] < L(v2)$

$0+4=4 < \infty$

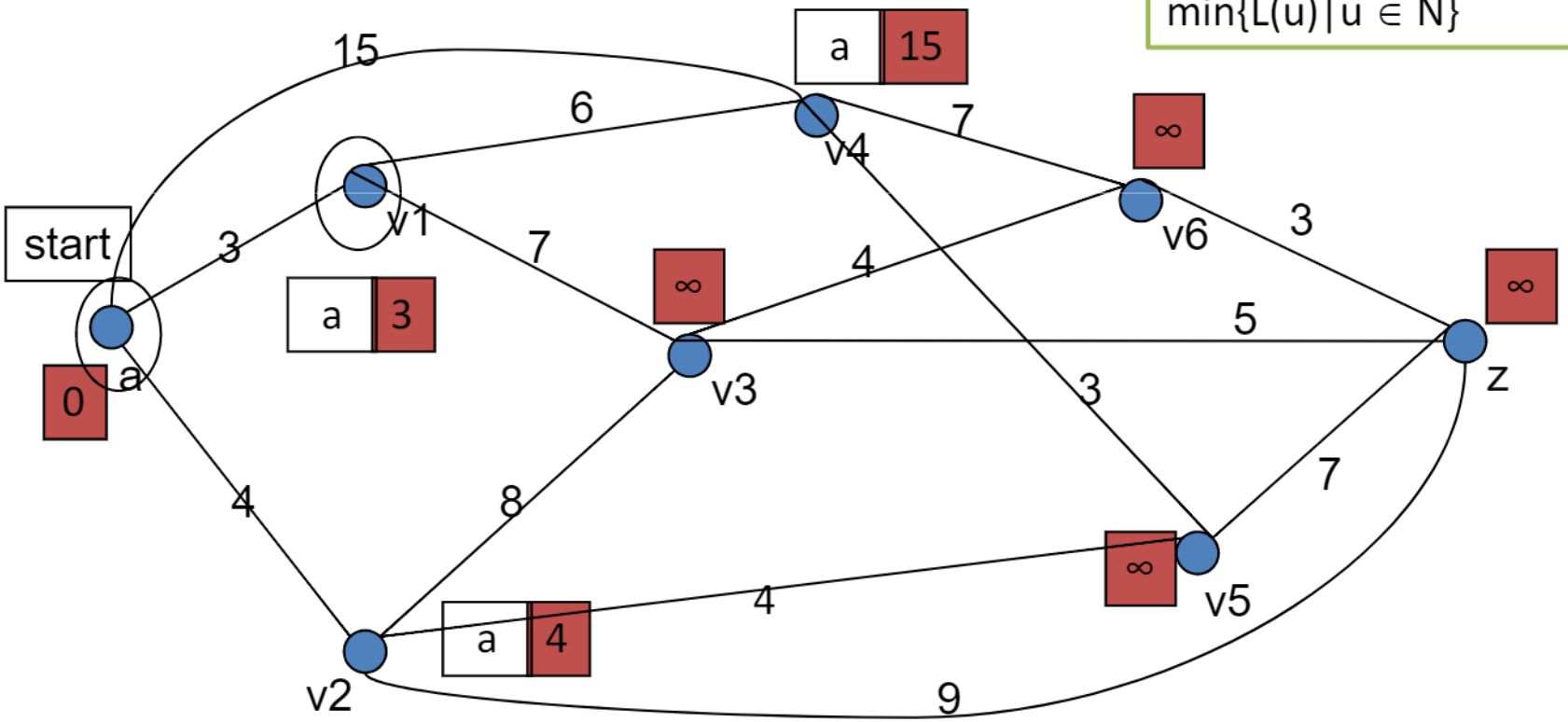
$L(v2)=4$



$S=\{a\}$

$N=\{v1,v2,v3,v4,v5,v6,z\}$

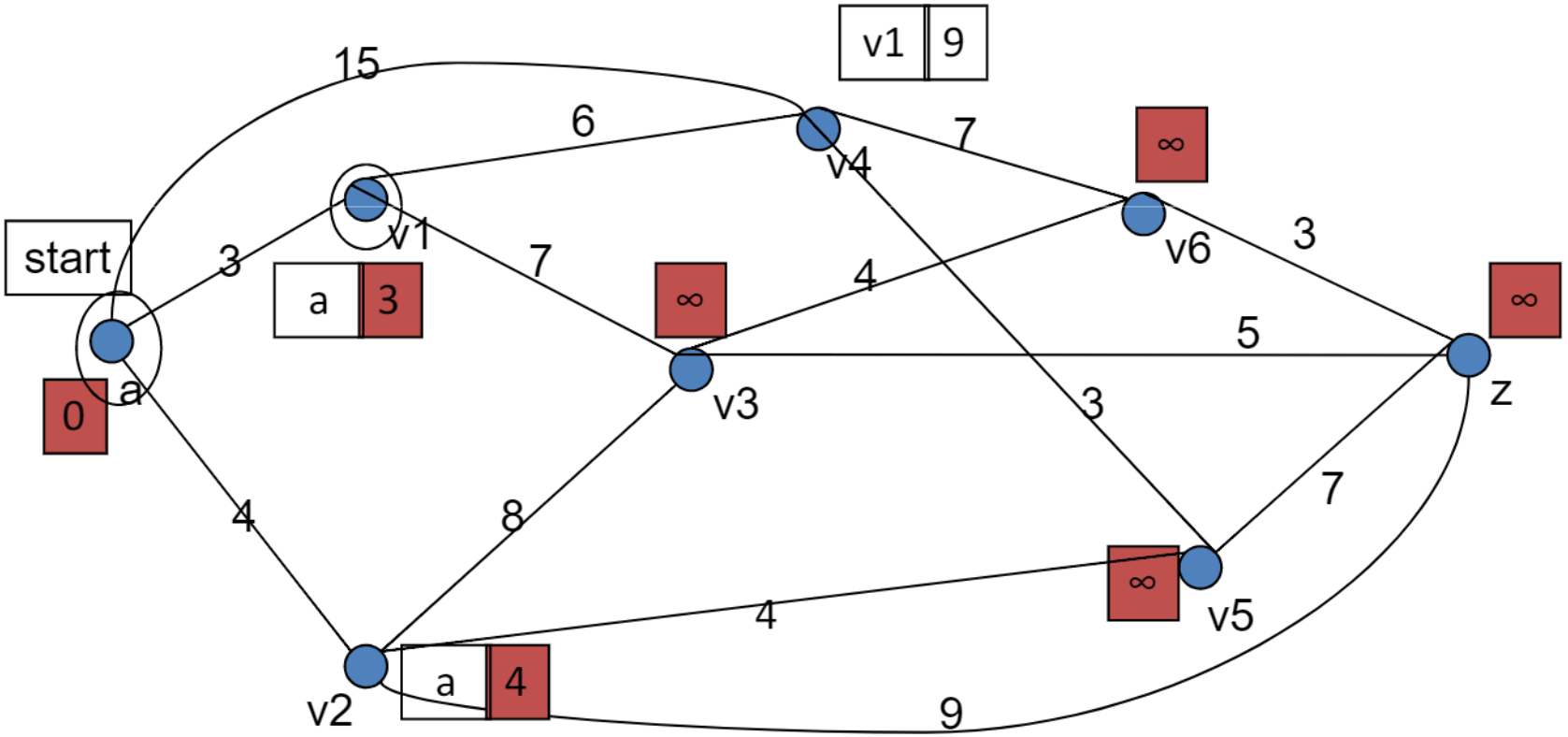
choose $v1$
because
$L(v1)=3$ =
$\min\{L(u) \mid u \in N\}$



$S=\{a, v1\}$

$N=\{v2,v3,v4,v5,v6,z\}$

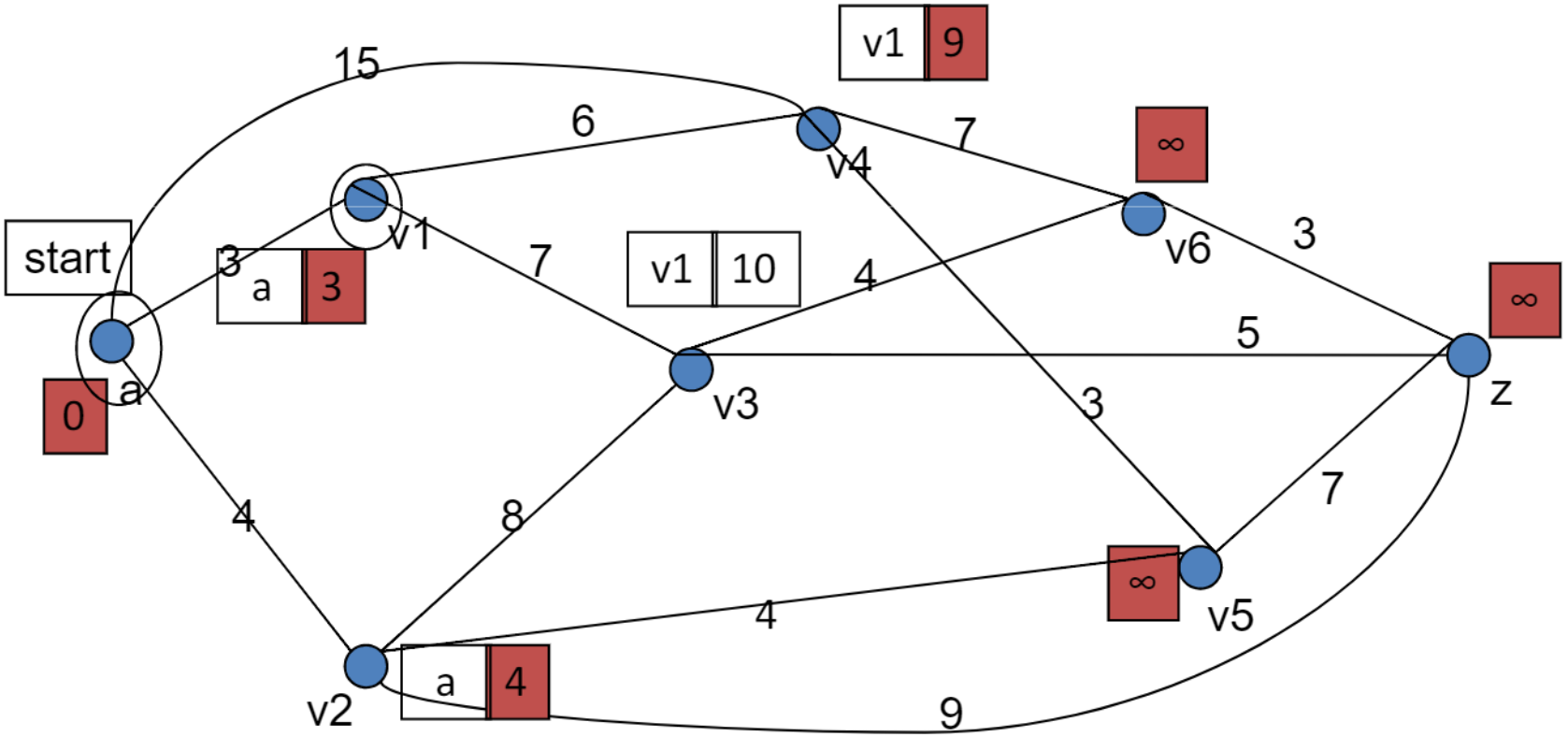
$L(v1)+W[v1,v4]<L(v4)$
$3+6= 9 < 15$
$L(v4)=9$



$S=\{a, v1\}$

$N=\{v2,v3,v4,v5,v6,z\}$

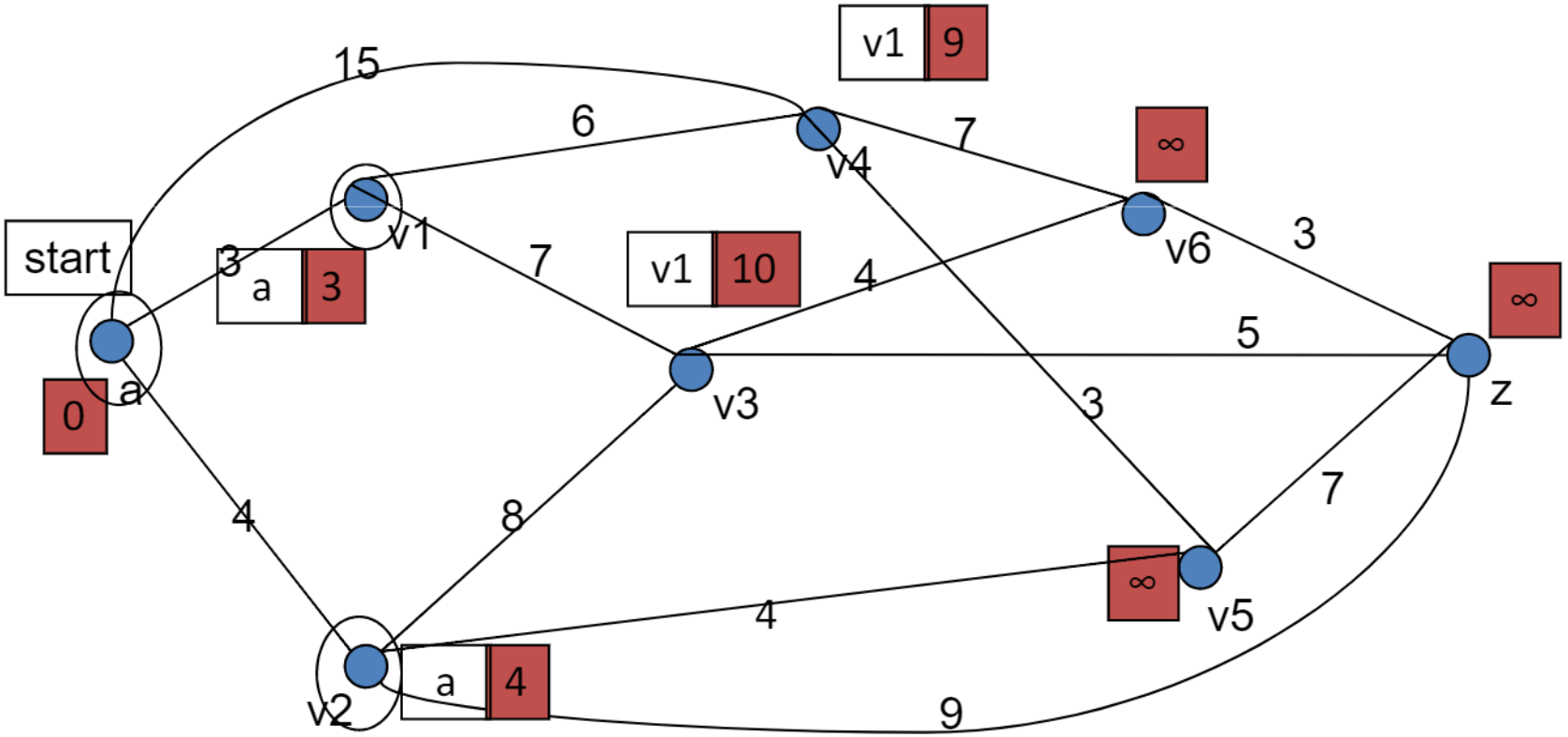
$L(v1)+W[v1,v3]<L(v3)$
$3+7=10<\infty$
$L(v4)=10$



$S=\{a, v1\}$

$N= \{v2,v3,v4,v5,v6,z\}$

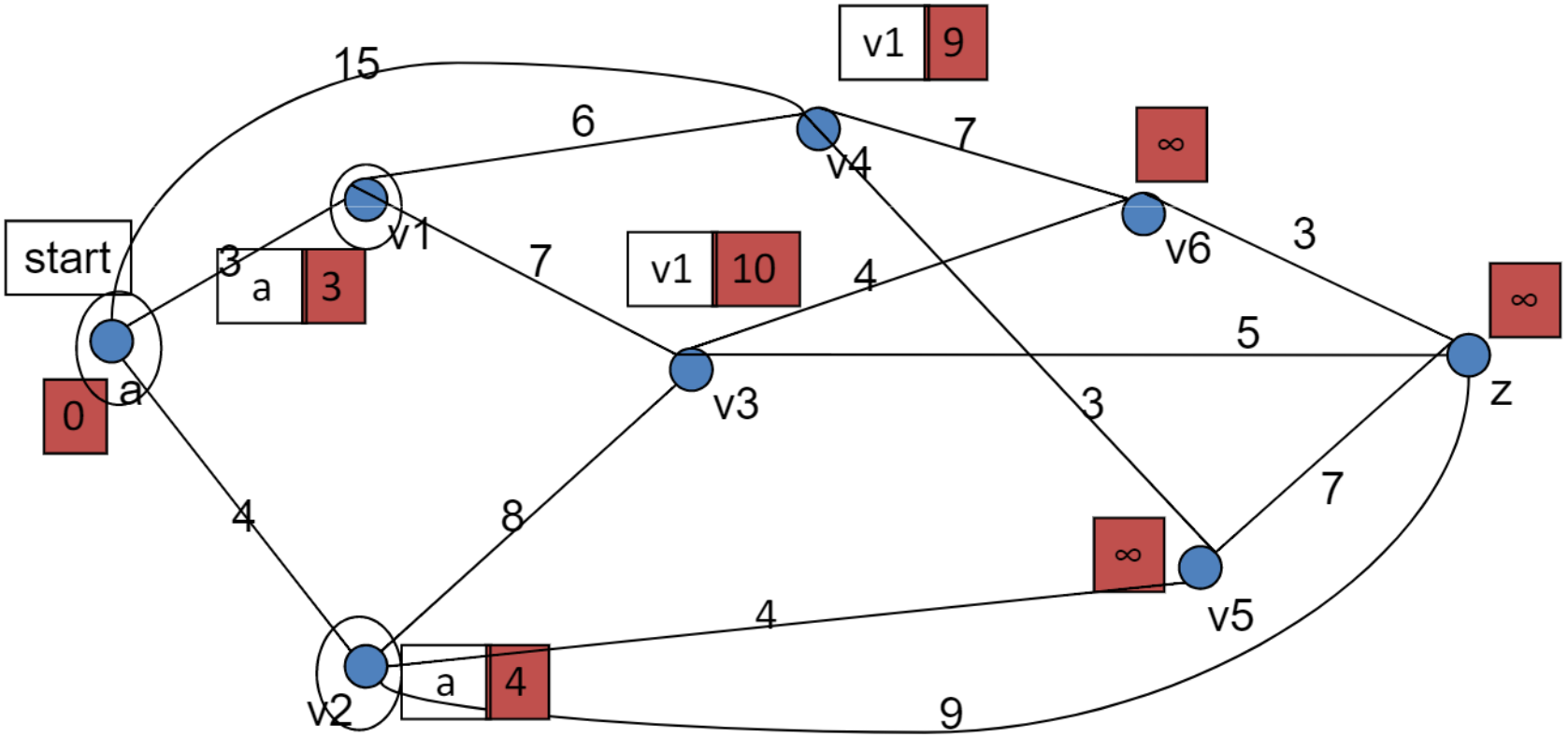
choose $v2$
because
$L(v2)=4 = \min\{L(u) \mid u \in N\}$



$S=\{a, v1, v2\}$

$N= \{v3,v4,v5,v6,z\}$

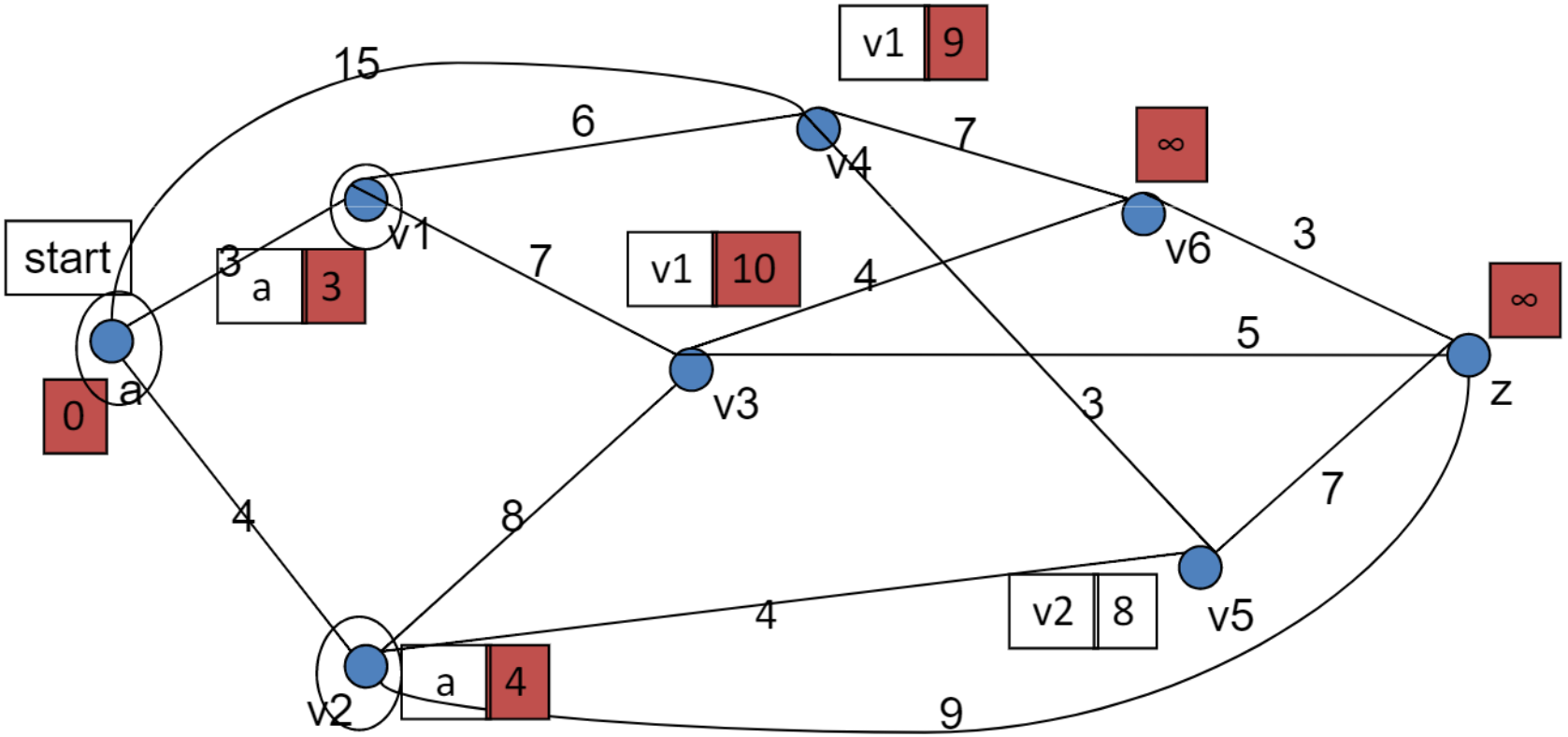
$L(v2)+W[v2,v3]<L(v3)$
$4+8= 12 > 10$
$L(v3)$ remains the same.



$S=\{a, v1, v2\}$

$N= \{v3,v4,v5,v6,z\}$

$L(v2)+W[v2,v5]<L(v5)$
$4+4= 8 < \infty$
$L(v5)=8$



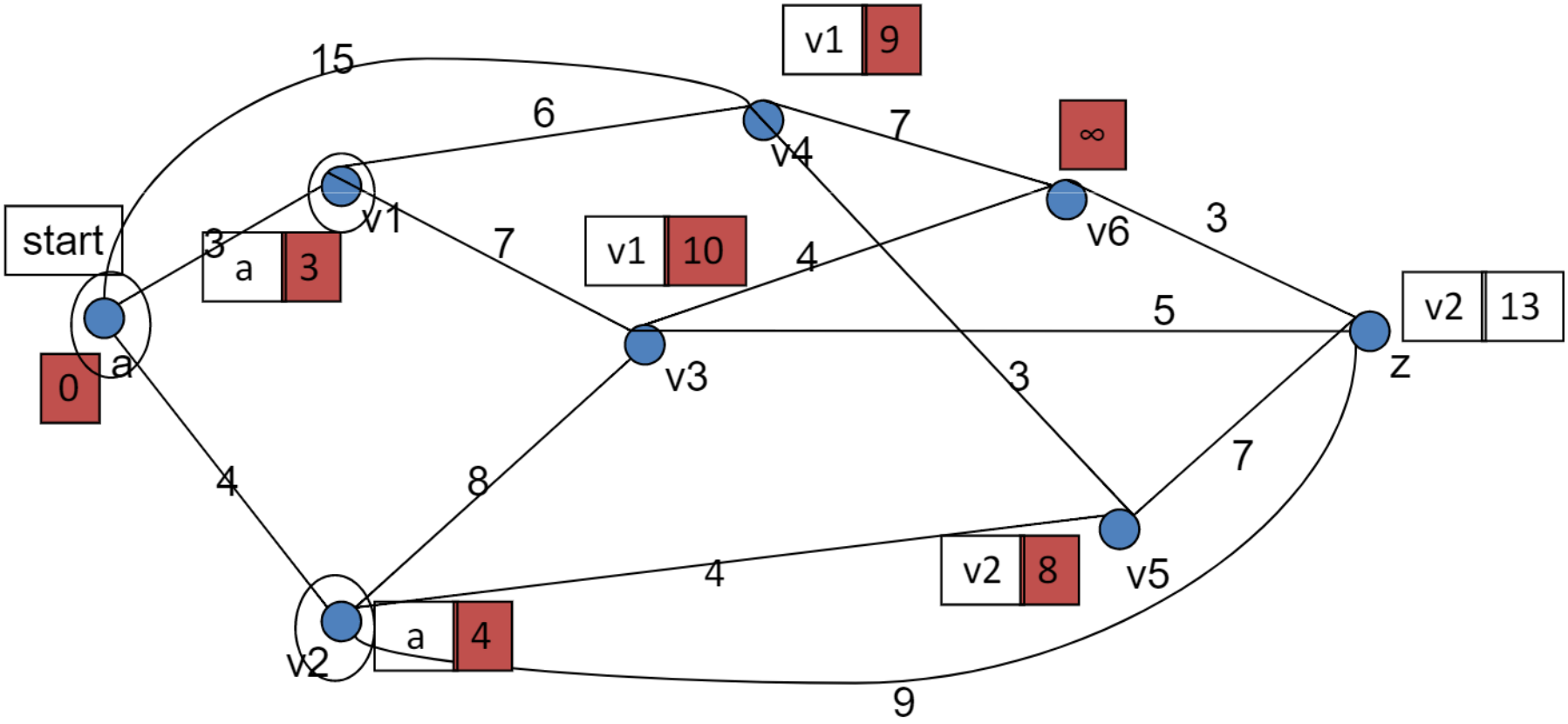
$S=\{a, v1, v2\}$

$N= \{v3,v4,v5,v6,z\}$

$L(v2)+W[v2, z]<L(z)$

$4+9= 13 < \infty$

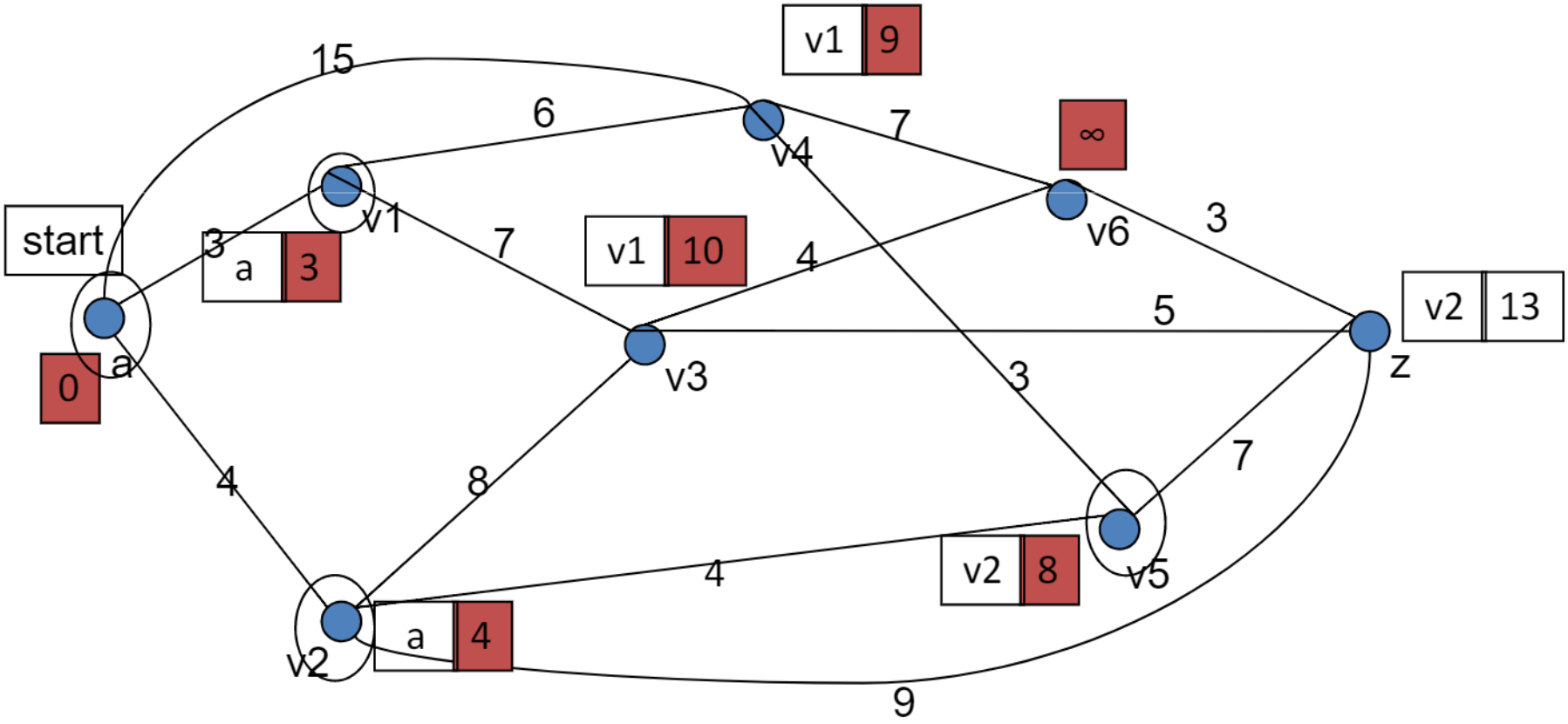
$L(z)=13$



$S=\{a, v1, v2\}$

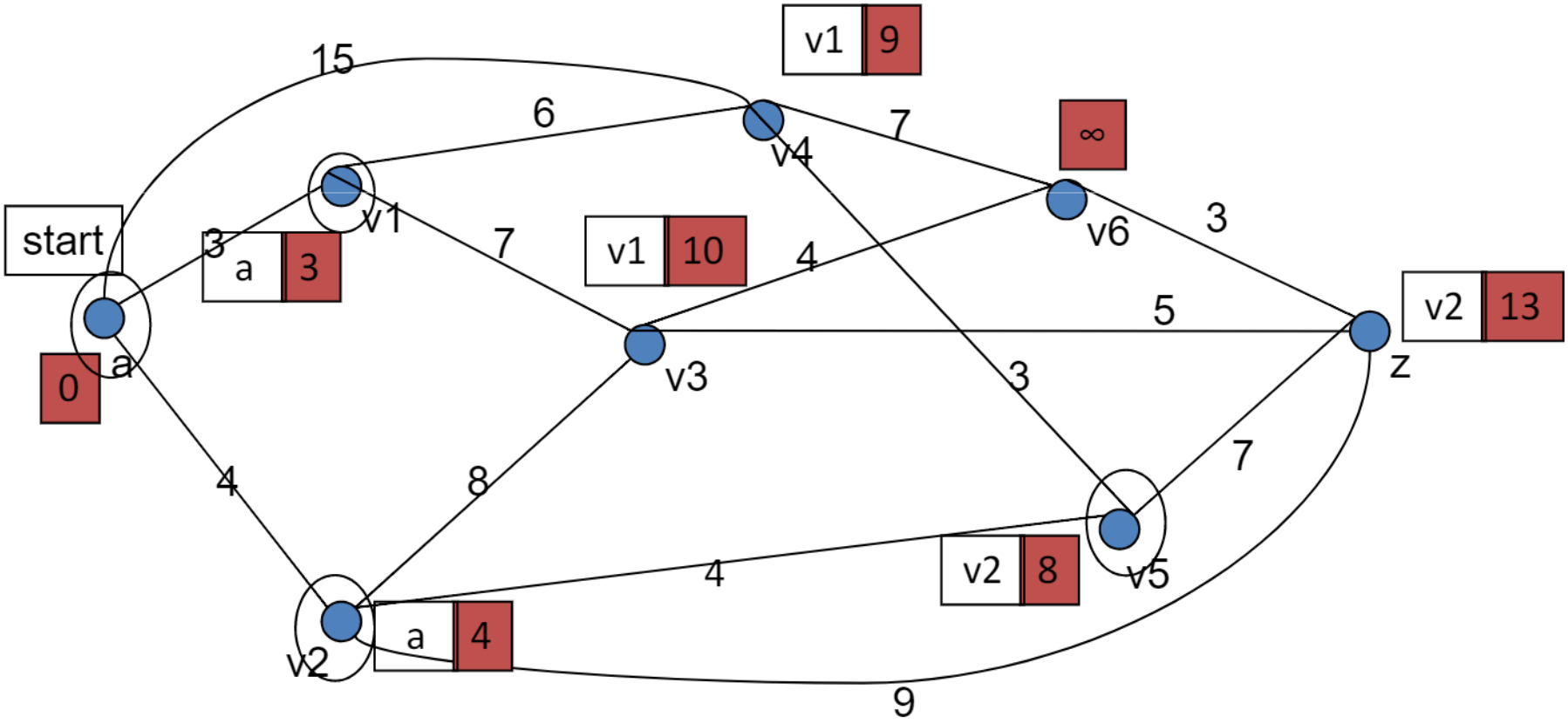
$N=\{v3,v4,v5,v6,z\}$

choose $v5$
because
$L(v5)=8 = \min\{L(u) \mid u \in N\}$



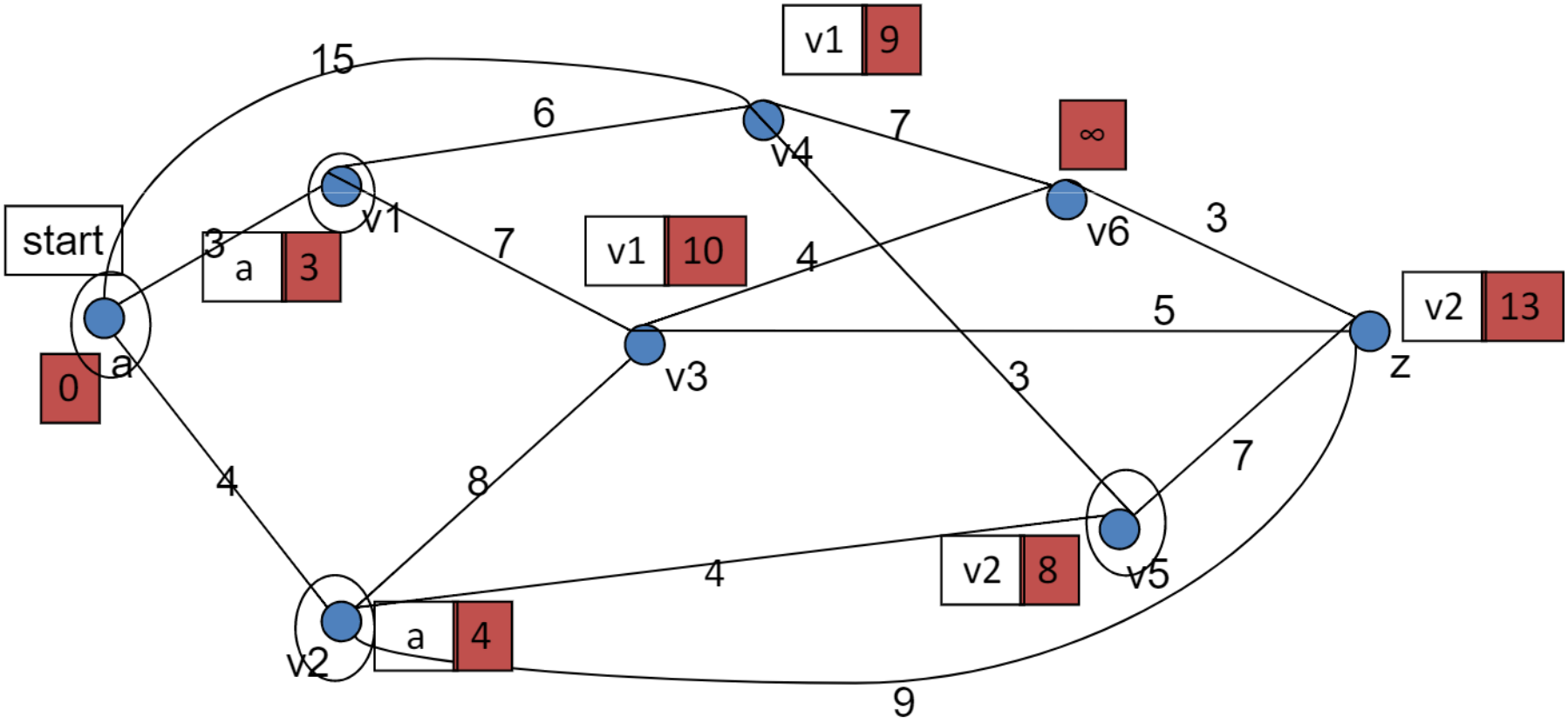
$S=\{a, v1, v2, v5\}$
$N=\{v3, v4, v6, z\}$

$L(v5)+W[v5, v4]<L(v4)$
$8+3= 11 > 9$
$L(v4)$ remains the same



$S=\{a, v1, v2, v5\}$
$N=\{v3, v4, v6, z\}$

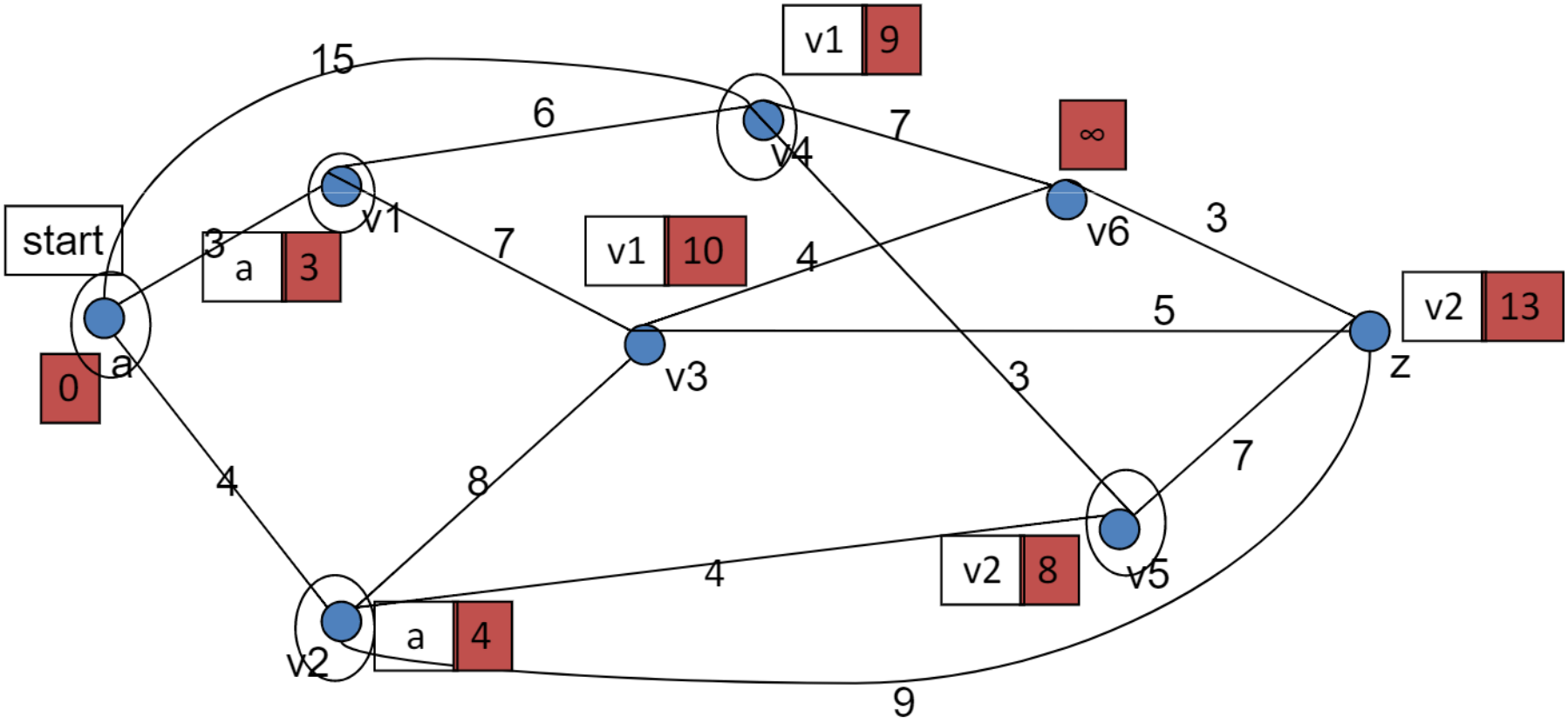
$L(v5)+W[v5, z] < L(z)$
$8+7= 15 > 13$
$L(z)$ remains the same



$S=\{a, v1, v2, v5\}$

$N=\{v3, v4, v6, z\}$

choose *v*4
because
$L(v4)=9 = \min\{L(u) \mid u \in N\}$



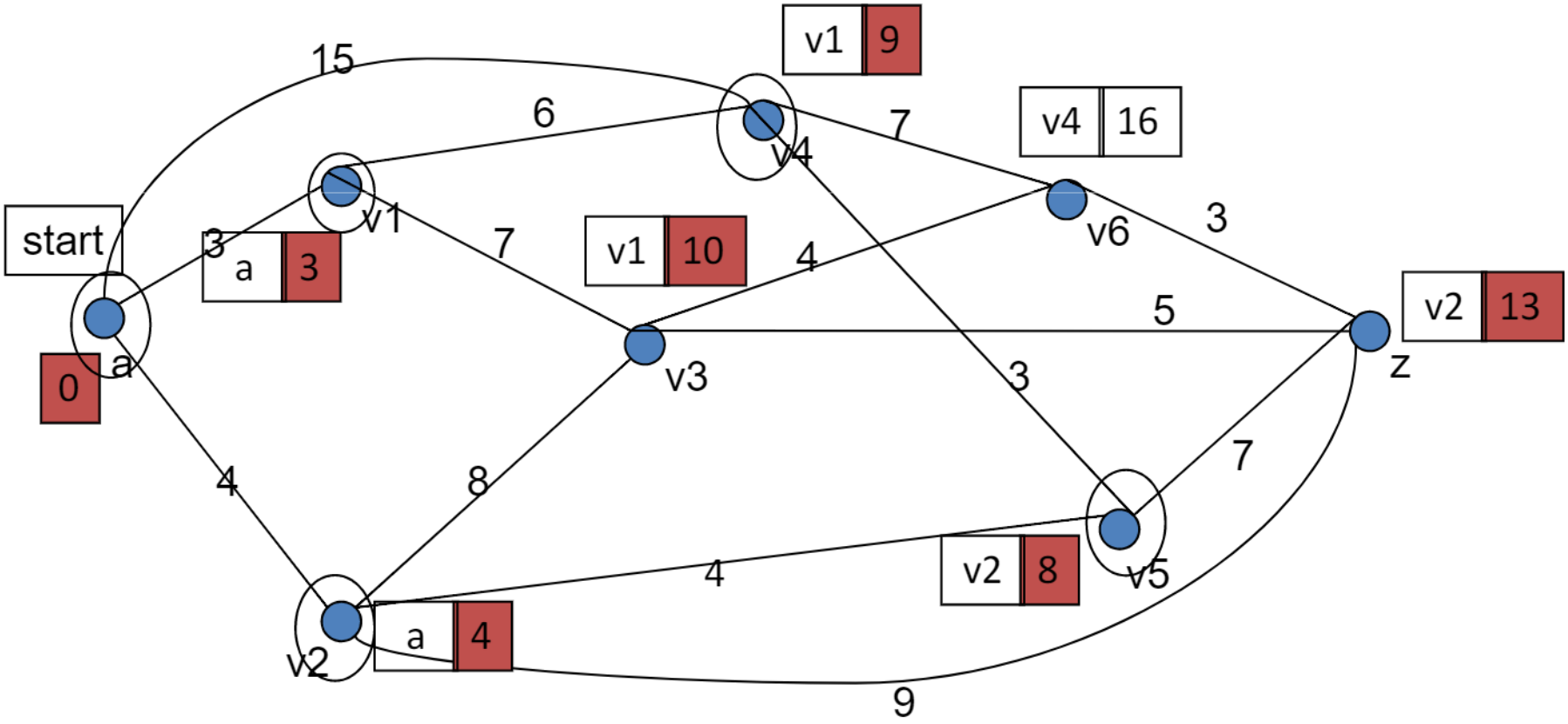
$S=\{a, v1, v2, v5, v4\}$

$N=\{v3, v6, z\}$

$L(v4)+W[v4, v6]<L(v6)$

$9+7= 16 < \infty$

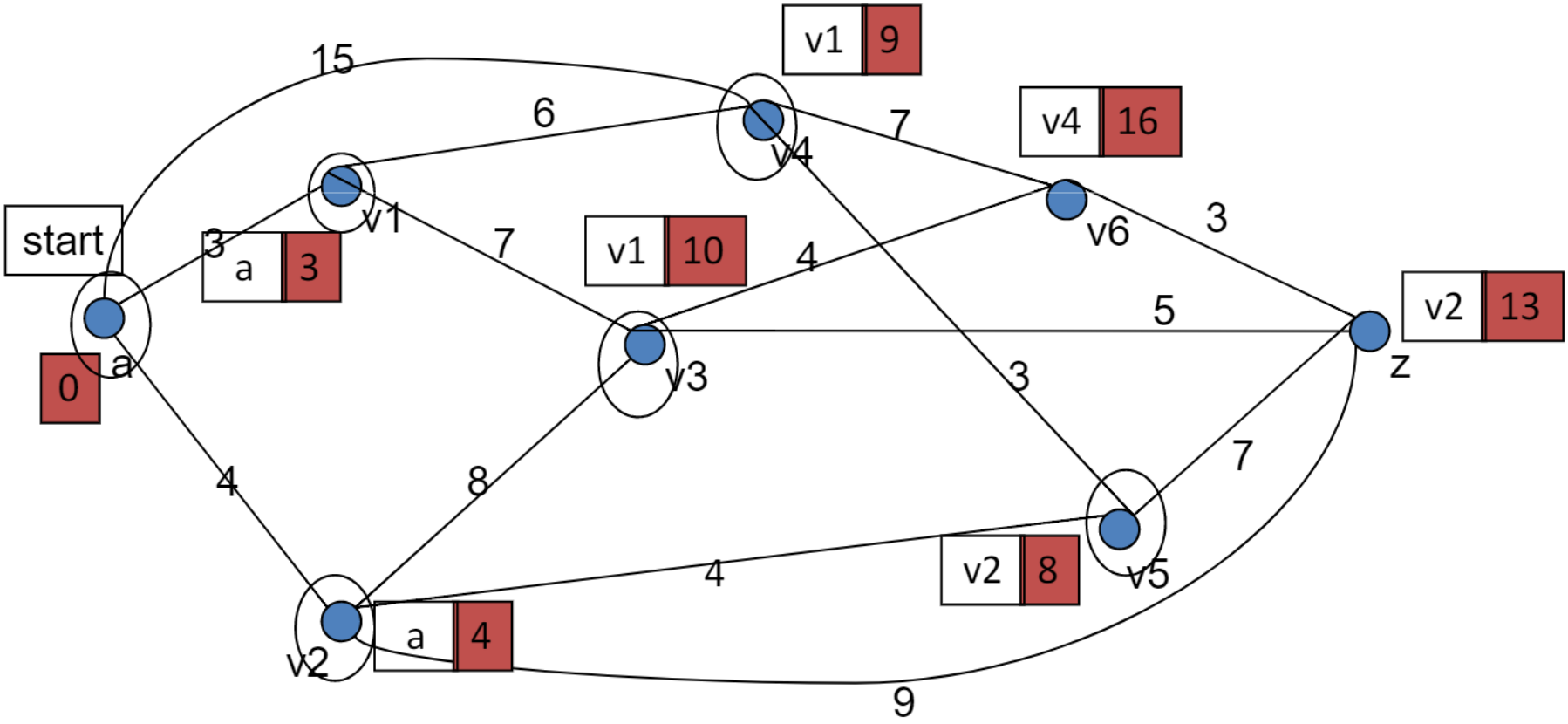
$L(v6) = 16$



S={a, v1, v2, v5,v4}

N= {v3, v6, z}

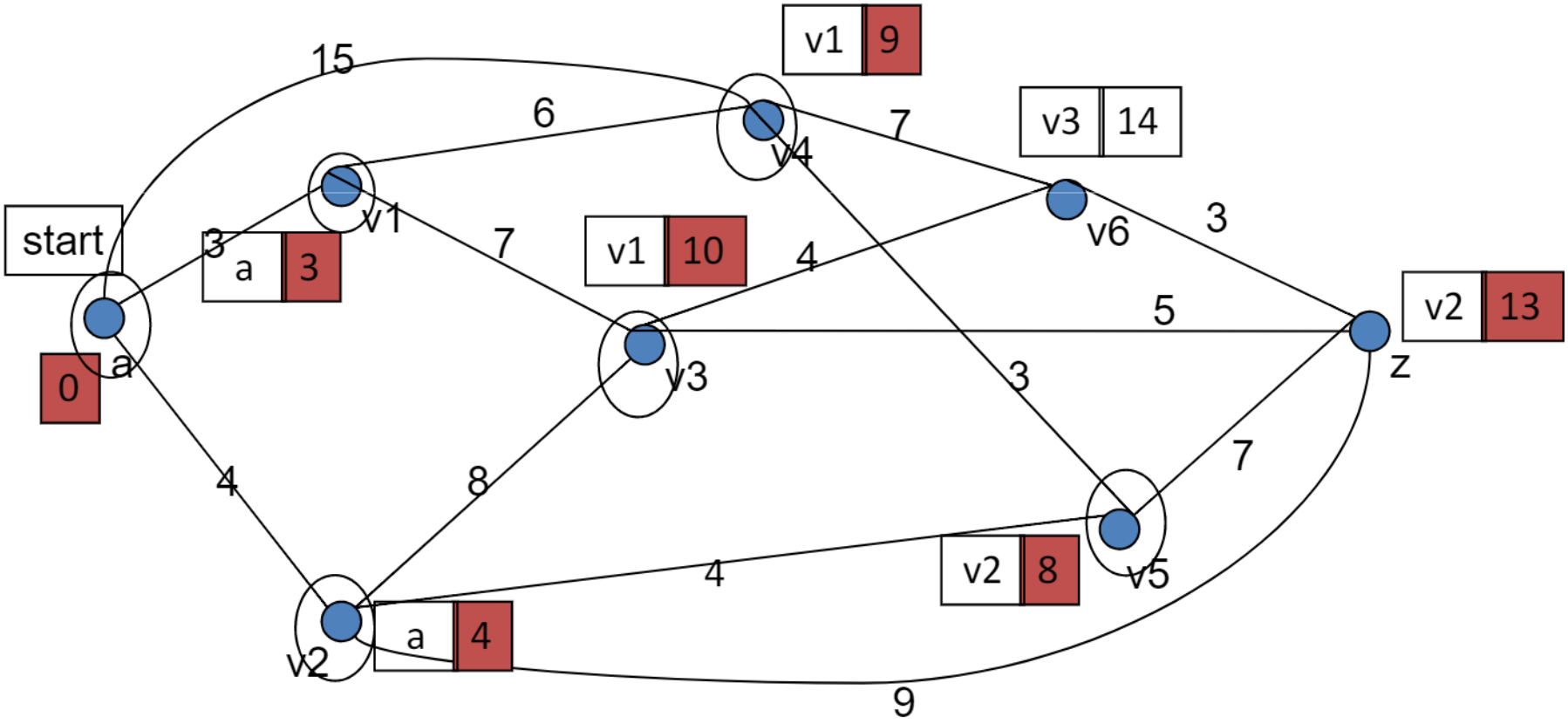
choose *v*3
because
$L(v3)=10 = \min\{L(u) \mid u \in N\}$



$S=\{a, v1, v2, v5, v4, v3\}$

$N=\{v6, z\}$

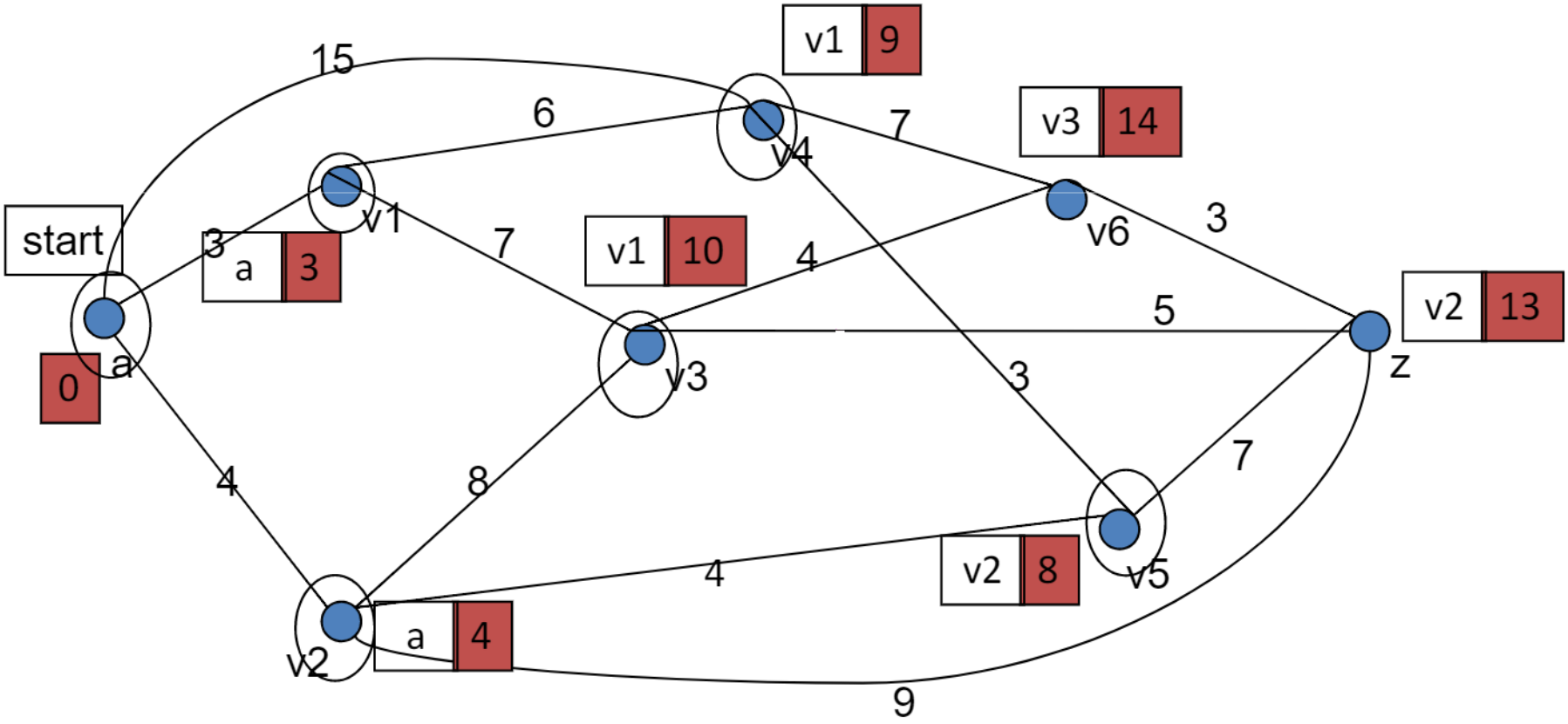
$L(v3)+W[v3, v6]<L(v6)$
$10+4= 14 < 16$
$L(v6) = 14$



$S=\{a, v1, v2, v5, v4, v3\}$

$N=\{v6, z\}$

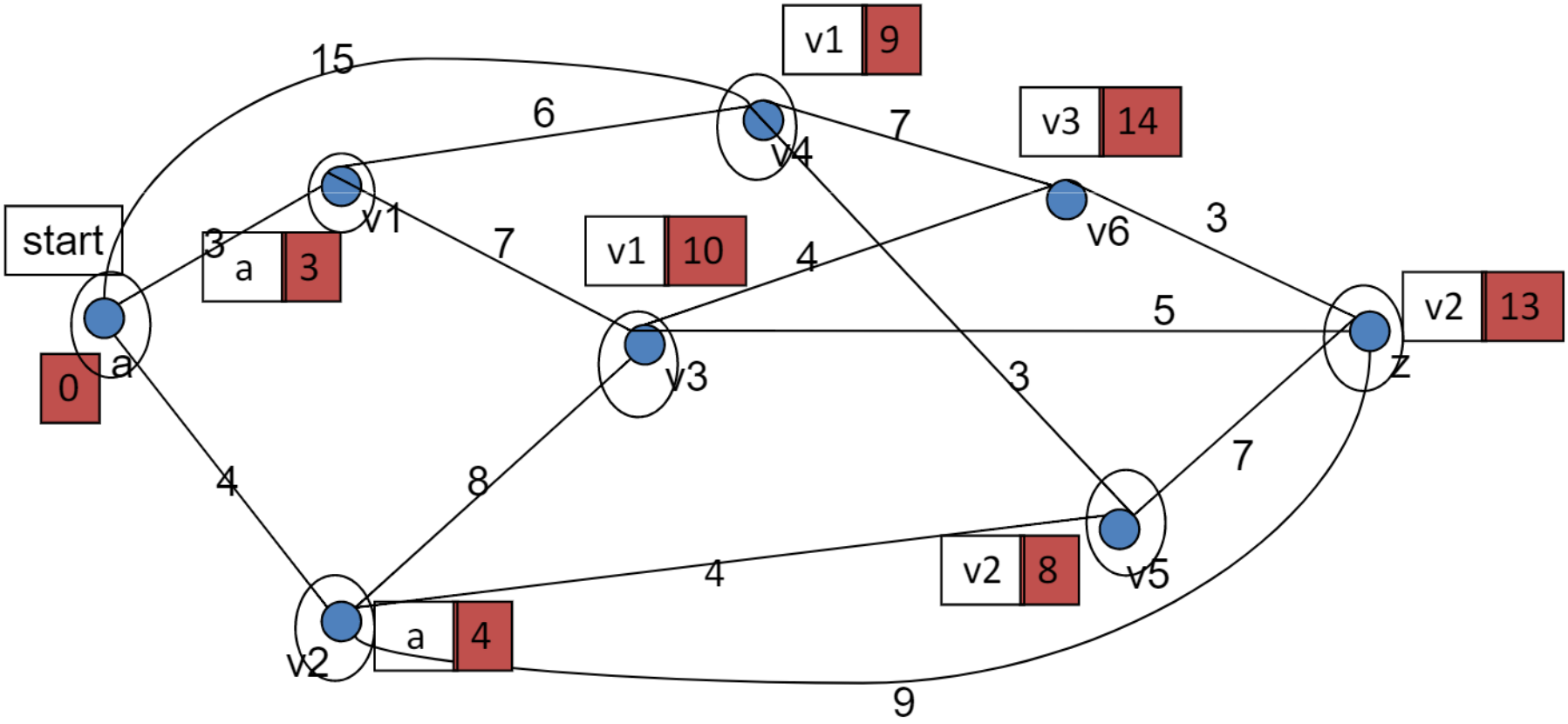
$L(v3)+W[v3, z] < L(z)$
$10+5= 15 > 13$
$L(z)$ remains the same



$S=\{a, v1, v2, v5, v4, v3\}$

$N=\{v6, z\}$

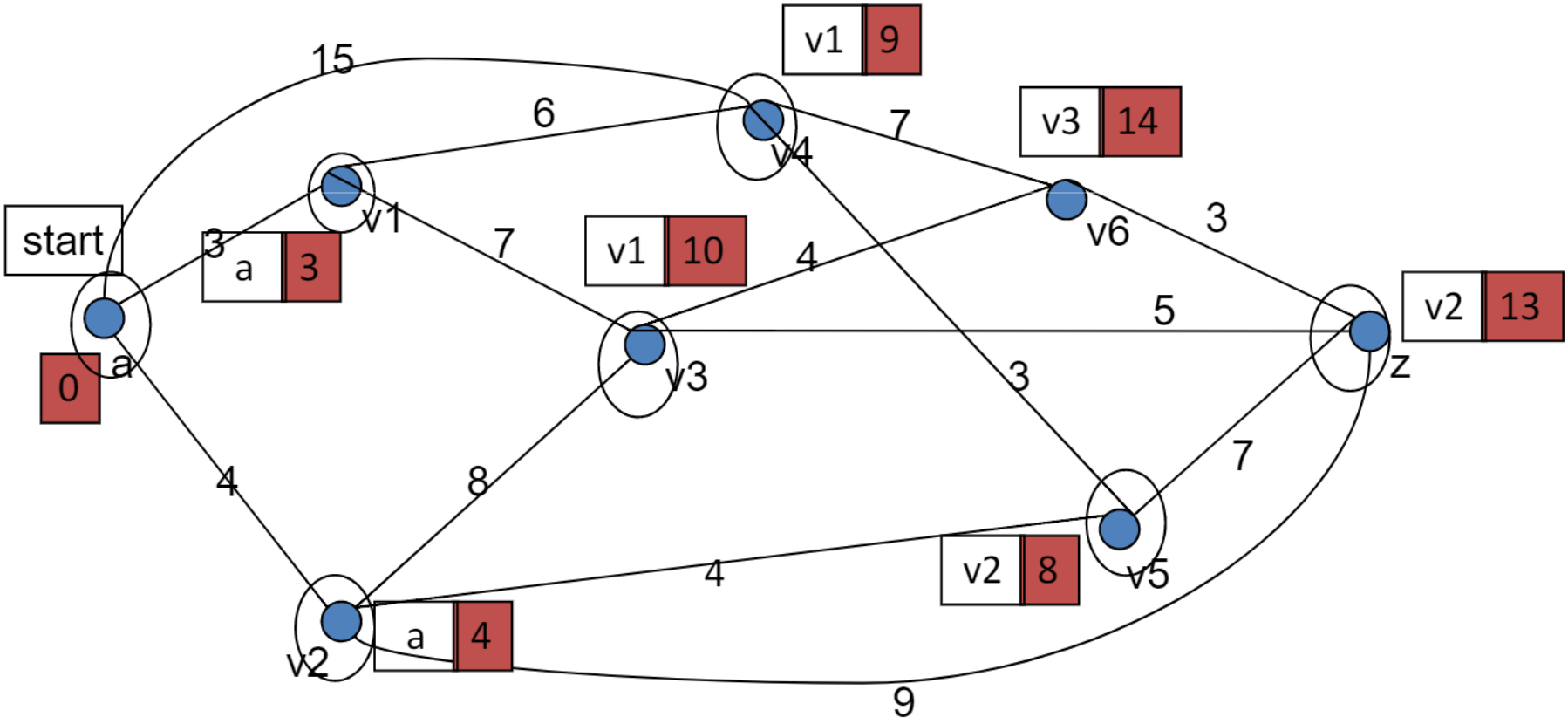
choose *z*
because
$L(z)=13 = \min\{L(u) \mid u \in N\}$



$S=\{a, v1, v2, v5, v4, v3, z\}$

$N=\{v6\}$

The loop terminates because $z \in S$



Shortest path from *a* to z is a -> v2-> z, with the length 13.

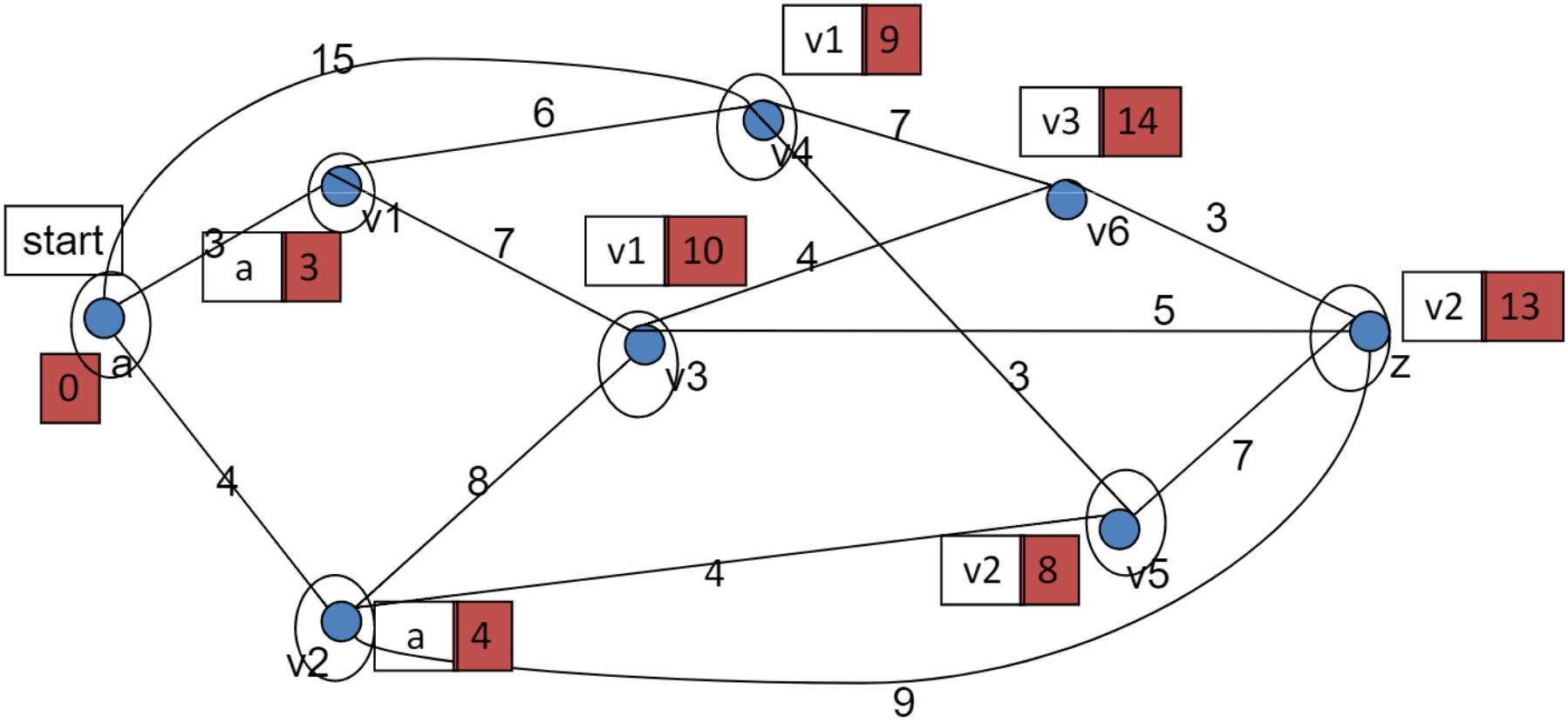
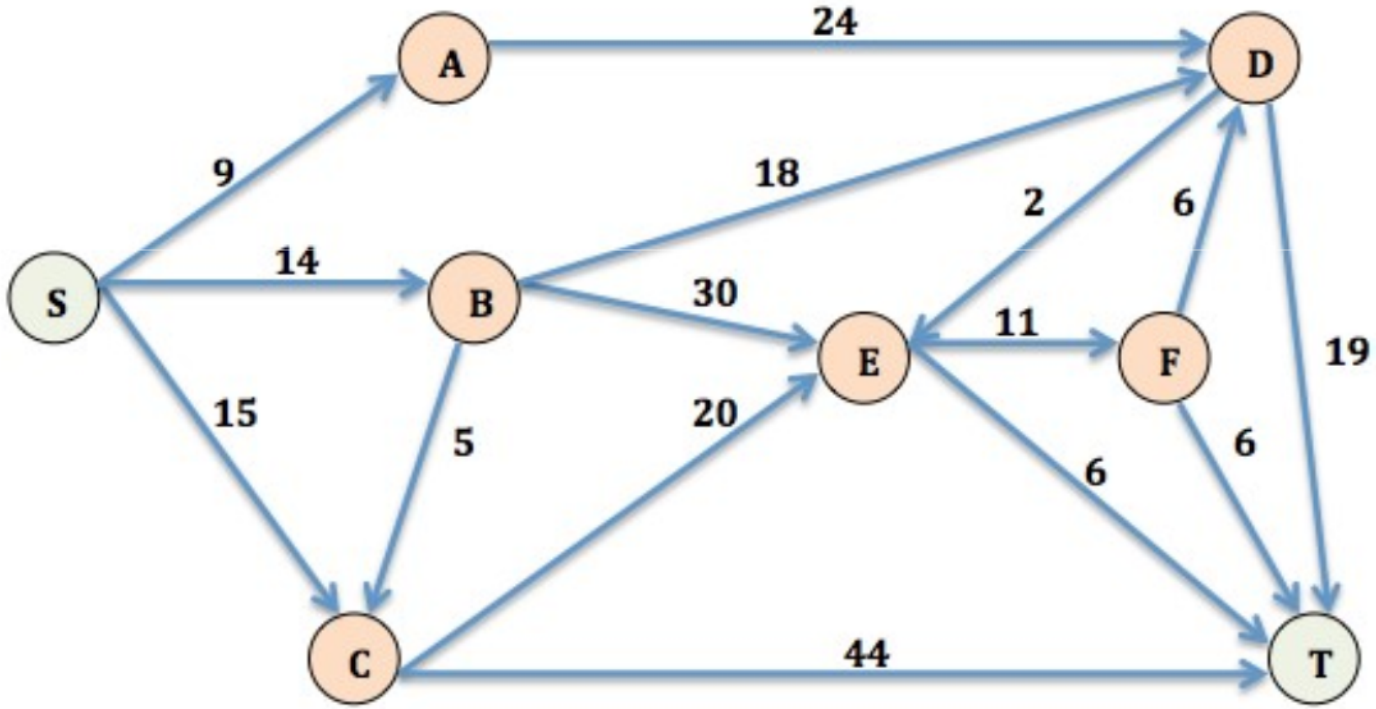


Table – Djikstra Algorithm

No.	S	N	L(a)	$L(V_1)$	$L(V_2)$	$L(V_3)$	$L(V_4)$	$L(V_5)$	$L(V_6)$	L(z)
0	{ }	{a, V_1, V_2, V_3, V_4, V_5, V_6, z }	0	∞	∞	∞	∞	∞	∞	∞
1	{a}	{V_1, V_2, V_3, V_4, V_5, V_6, z }		3	4	∞	15	∞	∞	∞
2	{a, V_1}	{V_2, V_3, V_4, V_5, V_6, z }		**3**	4	10	9	∞	∞	∞
3	{a, V_1, V_2}	{V_3, V_4, V_5, V_6, z }			**4**	10	9	8	∞	13
4	{a, V_1, V_2, V_5}	{V_3, V_4, V_6, z }				10	9	**8**	∞	13
5	{a, V_1, V_2, V_5, V_4}	{V_6, z }				10	**9**		16	13
6	{a, V_1, V_2, V_5, V_4, V_3,}	{V_6, z }				**10**			14	13
7	{a, V_1, V_2, V_5, V_4, V_3, z}	{V_6 }							14	**13**

Exercise

Q: Given a weighted digraph, find the shortest path from **S** to **T**, using Djikstra Algorithm.



Note: Weights are arbitrary numbers (i.e., not necessarily distances).